

Rot13

13 Key 25 و 26

- * Rotate by 13 places
- * Substitution cipher "replace each letter with 13 positions down the alphabet"
- * if you shift 13 positions again, you will back to original text
- * affects letters only (A-Z)
- * Same process for encryption and decryption

example 1

A	B	C	D	E	F	G	H	I	J	K	L	M
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

plain text: Hello

H → U

E → R

L → Y

L → Y

O → B

cipher text: URyyB

example 2

plaintext: Hello 123!

H → U

E → R

L → Y

L → Y

O → B

cipher text: URyyB 123!

example 3

plaintext: Hide The Message

H → U

i → V

d → Q

E → R

T → G

h → u

e → R

M → Z

e → R

S → F

S → F

a → N

g → T

e → R

cipher text: UVQR GUR ZRFFNTR

* Numbers and spaces and special chars don't change

cipher text: URyyB 123!

OTP:

and auto generated

- * One-time pad
- * Perfect encryption Scheme, ultra Secure low bandwidth in 1917
- * Key is a random sequence of the same length as the message
- * Add the Key to the message bit by bit (XOR operation)
- * Discard the Key and never use again
by both Sender and Receiver
- * Two copies of Key: one with sender and other with receiver
- * Decryption uses the same key but Subtracts instead of adding
- * Encryption is unbreakable for a ciphertext only Attack
- * Key is Random sequence of Shift between 0 and 25

~~if west~~
if west
From 0
Start From 1

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
if west	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Start From 1	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26

Example 1

plaintext: ONETIMEPAD
Key: TBF RGF ARFM

$$\begin{aligned}
 O(15) + T(20) &= 35 \% 26 = 9 \text{ (I)} \\
 N(14) + B(2) &= 16 \% 26 = 16 \text{ (P)} \\
 E(5) + F(6) &= 11 \% 26 = 11 \text{ (K)} \\
 T(20) + R(18) &= 38 \% 26 = 12 \text{ (L)} \\
 I(9) + G(7) &= 16 \% 26 = 16 \text{ (P)} \\
 M(13) + F(6) &= 19 \% 26 = 19 \text{ (S)} \\
 E(5) + A(1) &= 6 \% 26 = 6 \text{ (F)} \\
 P(16) + R(18) &= 34 \% 26 = 8 \text{ (H)} \\
 A(1) + F(6) &= 7 \% 26 = 7 \text{ (G)} \\
 D(4) + M(13) &= 17 \% 26 = 17 \text{ (Q)}
 \end{aligned}$$

ciphertext = IPKLP SFHGQ

Example 2

plaintext: 00101001
Key: 10101100

$$\begin{array}{r}
 00101001 \\
 + 10101100 \\
 \hline
 10000101
 \end{array}$$

XOR = 2

ciphertext

Caesar cipher

لوجان ال Key عبارة عن حرف حوله رقم
لوجان ال Key بالاسان او الكرمي 25 طيب قاعد 26 %

- * Mono-alphabetic Substitution - Julius Caesar «communicate with genera
- * Shifted fixed number of places down the alphabet
- * Symmetric encryption - Decryption is inverse of encryption
- * Transformation letters into numbers «modular Arithmetic»

$$\text{Encryption: } E_n = (X + n) \% 26$$

$$\text{Decryption: } D_n = (X - n) \% 26$$

* Exhaustive Key Search (brute force Attack) is Available because the key space is too small 26^{pos} ke

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25

example 1

plaintext = Send Secret Key

$$K = 16$$

secret key table

A	B	C	D	E	F	G	H	I	J	K	L	M
Q	R	S	T	U	V	W	X	Y	Z	A	B	C
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
D	E	F	G	H	I	J	K	L	M	N	O	P

S → i

e → u

n → D

d → t

s → i

e → u

c → s

r → h

e → u

t → j

k → A

e → u

y → o

Ciphertext: iudt iushuj Auo!

example 2

ciphertext = hp htwx Xppe

$$K = 11$$

secret key table

A	B	C	D	E	F	G	H	I	J	K	L	M
P	Q	R	S	T	U	V	W	X	Y	Z	A	B
n	o	p	q	r	s	t	u	v	w	x	y	z
C	D	E	F	G	H	I	J	K	L	M	N	O

h → w

p → e

h → w

t → i

w → L

w → L

x → m

p → e

p → e

e → t

plaintext = we will meet

Row transposition

* if Key give in letters, convert it into numbers based on the Rank

Example → Key: Zebras → convert into → Key: 632415

Steps:

Encryption { * نكتب ال Key زي مافو من الشمال لليمين بحديت برون تحت ال plaintext وبمنى Row by Row
* نجمع الحروف بترتيب الارقام بحيث تبقى column by column

Decryption { * نكتب ال Key زي مافو من الشمال لليمين بحديت برون تحت ال cipher text وبمنى column by column ونأبرهن حسب الترتيب
* نجمع ~~الحروف~~ وبمنى Row by Row

Example 1

Plaintext Attack post pone duntill
two amxy z
Key → 4312567

4	3	1	2	5	6	7
A	t	t	a	c	k	p
o	s	t	p	o	n	e
d	u	n	t	i	l	t
w	o	a	m	x	y	z

cipher text:

ttna aptm tsuo
aodw coix knly
petz

Example 2

ciphertext : lo ewd hol Lr
Key : 3214

3	2	1	4
h	e	L	L
o	w	o	r
L	d		

Cipher text:

Hello world

Playfair :-

- 1) 5x5 matrix without reaption characters حتى لو كان التكرار في اد Key نفس
- 2) Same row $\rightarrow$ Shift ~~right~~ Decryption $\rightarrow$ Shift ~~left~~
- 3) Same column $\rightarrow$ Shift down Decryption $\rightarrow$ Shift up
- 4) Different $\rightarrow$ (interSection in row)
- 5) if odd $\rightarrow$ put X or Z
- 6) if similar letter inside of each other, put X between them put i-j together in matrix

Example 1

plaintext : Balloon

Key : monarchy

Ba lx lo on
m o n a r
c h y b d
e f g i j k
l p q s t
u v w x z

ciphertext :

ib Su pm na

Example 2

ciphertext : g d d o g d r q p r s d h m e m b v

Key : charles

~~c h a r l~~
~~e s b d f~~
~~g h i j k m~~
~~n o p q t~~

c h a r l
e s b d f
g i j k m n
o p q t u
v w x y z

plaintext : meet me at the bridge X

I will ignore it because it additional letter

Example 3

ciphertext : L F G D N W D p w o c v

Key : Hello world

~~H e l o w~~
~~r d b c f~~
~~g i j k m n~~

H e l o w
r d a b c
f g i j k m
n p q s t
u v x y z

plaintext : Hide the gold Z

T. will ignore it - Additional letter

At-Fine:-

Encryption $\rightarrow C = (aX + B) \text{ Mod } 26$

Decryption $\rightarrow X = a^{-1}(c-b)$

Key $\rightarrow (a, b)$

Example 1

Message = 90
Key = $(7, 2)$

$$C = (aX + b) \% 26 = (6 \times 7 + 2) \% 26 \\ = (42 + 2) \% 26 = 44 \% 26 = 18 (S)$$

$$C = (14 \times 7 + 2) \% 26 = (98 + 2) \% 26 = \\ 100 \% 26 = 22 (w)$$

Ciphertext: sw

Example 2

Cipher = sw²²
Key = $(7, 2)$

$$X = a^{-1}(c-b) = \\ (7^{-1})(18-2)$$

$$a^{-1} = 15 \times 7 \% 26 = 1$$

$$X = 15(18-2) = 15 \times 16 \% 26 = \\ 240 \% 26 = 6 (G)$$

$$X = 15(22-2) = 15 \times 20 \% 26 = \\ 300 \% 26 = 14 (o)$$

plaintext: Go

Example 3

Message: Sending secret message

Key: $(17, 13)$
a b

$$C = (aX + b) \% 26$$

S(18) $\rightarrow (17 \times 18 + 13) \% 26 = 7 (H)$
e(4) $\rightarrow (17 \times 4 + 13) \% 26 = 3 (D)$
n(13) $\rightarrow (17 \times 13 + 13) \% 26 = 0 (A)$
d(3) $\rightarrow (17 \times 3 + 13) \% 26 = 12 (M)$
i(8) $\rightarrow (17 \times 8 + 13) \% 26 = 19 (T)$
n(13) $\rightarrow (17 \times 13 + 13) \% 26 = 0 (A)$
g(6) $\rightarrow (17 \times 6 + 13) \% 26 = 11 (L)$

and so on

Ciphertext: Hdamtal HdVqdy
Jdhhnld

c(2) $\rightarrow (17 \times 2 + 13) \% 26 = 21 (V)$
r(17) $\rightarrow (17 \times 17 + 13) \% 26 = 16 (Q)$
t(9) $\rightarrow (17 \times 9 + 13) \% 26 = 24 (Y)$
m(12) $\rightarrow (17 \times 12 + 13) \% 26 = 9 (J)$
a(0) $\rightarrow (17 \times 0 + 13) \% 26 = 13 (N)$

Example u

Ciphertext : ZS gms

plaintext : pe

Key ?

rest of plaintext ?

$$C = ax + b$$

$$\begin{aligned} 25 &= 15a + b \\ 18 &= 4a + b \end{aligned} \quad \text{نطرح المعادلتين}$$

$$7 = 11a$$

$$a = 7 \times 11^{-1}$$

$$a = 7 \times 19 = 133 \% 26 = 3$$

هنا نخرج زعوفنا في اي معادلة من التي فوق
علشان نجيب قيمة a

$$18 = 4 \times 3 + b$$

$$18 = 12 + b$$

$$b = 6$$

Key : (3, 6)

$$X = a^{-1} (b - c)$$

↓

$$3 \times 9 \% 26 = 1 \quad 3^{-1} = 9$$

هنا عادي لو بدلت
بصيت ناتج الطرح
صيرنا مايب

$$X = 9 (6 - 6) = 9 \times 0 = 0 \text{ (A)}$$

$$X = 9 (6 - 12) = 9 \times -6 = -54 \% 26 = 2 \text{ (C)}$$

$$X = 9 (6 - 18) = 9 \times -12 = -108 \% 26 = 4 \text{ (e)}$$

peace

Substitution:

* هنا اذا اللي بعمل Random Key بنفسي فبورها كل الحروف وتحتها يكتب الحرف اللي لهو مع
عنه بيه بب بترط أن مفيش حرف يتكرر

Example 1

plaintext: rootit

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
n	a	h	y	x	p	o	e	k	j	d	i	u	g	q	z	w	b	t	l	s	f	r	c	v	

Ciphertext: BqqLKL

* لو عاينر تعمل Decryption هتدوع لازم يبقى معاك نفس الجردو وتبدل الحروف

Vegenere:

$$C = P + K \% 26$$

$$P = C - K \% 26$$

* فكرته زي ال OTP بس الاختلاف أنه ال Key
مش بيبقى على قد الكلمة لا أنت بتفعل
تكرر فيه لحد ما تخفي كل ال plaintext الى عندك

Example 1

Message: hello Mark
Key: iteam

h	e	l	l	o	M	a	r	k	
+	i	t	e	a	m	i	t	e	a
7	4	11	11	14	12	0	17	10	
+	8	19	4	0	12	8	19	4	0
15	23	15	11	26	20	19	21	10	
after mod 26									
(15	23	15	11	0	20	19	21	10)	
(P	X	P	L	a	u	t	v	K)	

Ciphertext: pXPLautvK

Example 2

Message: we will meet
Key: one

w	e	w	i	l	l	m	e	e	t	
+	o	n	e	o	n	e	o	n	e	o
22	4	22	8	11	11	12	4	4	19	
+	14	13	4	14	13	4	14	13	4	14
36	17	26	22	24	15	26	17	8	33	
after mod 26										
(10	17	0	22	24	15	0	17	8	7)	
(K	r	a	w	y	P	a	r	i	h)	

Ciphertext: KrawyParih

fill 2x2 :-

Key ممکن ہے یا نہیں وہ ABCd فہرست میں ماتریکس بالمشق دہ

Example 1

$$\text{key} \rightarrow \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\text{plaintext} \rightarrow \overset{7}{h} \overset{19}{a} \overset{18}{b} s$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 7 \\ 0 \end{bmatrix} = \begin{bmatrix} 21 \\ 14 \end{bmatrix} = \begin{bmatrix} V \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 19 \\ 18 \end{bmatrix} = \begin{bmatrix} 111 \\ 128 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 7 \\ 24 \end{bmatrix} = \begin{bmatrix} H \\ y \end{bmatrix}$$

$$\text{ciphertext} \rightarrow Vohy$$

Example 2

$$\text{key} \rightarrow \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\text{ciphertext} \rightarrow \overset{21}{V} \overset{14}{o} \overset{7}{h} \overset{24}{y}$$

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 21 \\ 14 \end{bmatrix} = \begin{bmatrix} 553 \\ 546 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 7 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} h \\ a \end{bmatrix}$$

$$1) \text{determine} \rightarrow (3 \times 5) - (2 \times 3) = 15 - 6 = 9$$

$$2) \text{inverse of } d \rightarrow 3 \times 9 \% 26 = 1$$

$$\begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} 7 \\ 24 \end{bmatrix} = \begin{bmatrix} 513 \\ 356 \end{bmatrix} \rightarrow \begin{bmatrix} 19 \\ 18 \end{bmatrix}$$

$$\begin{bmatrix} t \\ s \end{bmatrix}$$

$$3) \text{key}^{-1} = \frac{1}{d} \text{adj}$$

$$\text{adj} = \begin{pmatrix} 5 & -3 \\ -2 & 3 \end{pmatrix} \xrightarrow{\text{mod } 26} \begin{pmatrix} 5 & 23 \\ 24 & 3 \end{pmatrix}$$

$$\text{key}^{-1} = 3 \times \begin{bmatrix} 5 & 23 \\ 24 & 3 \end{bmatrix} = \begin{bmatrix} 15 & 69 \\ 72 & 9 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix}$$

hill 3x3 :-

Example 1

key $\rightarrow$ G y b n q k u R p

plaintext $\rightarrow$ ^{0 2 19 8 13 6} Acting

$$\text{Key} \rightarrow \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} = \begin{bmatrix} 67 \\ 222 \\ 319 \end{bmatrix} \rightarrow \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix} = \begin{bmatrix} P \\ o \\ h \end{bmatrix}$$

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 8 \\ 13 \\ 6 \end{bmatrix} = \begin{bmatrix} 366 \\ 372 \\ 471 \end{bmatrix} \rightarrow \begin{bmatrix} 2 \\ 8 \\ 3 \end{bmatrix} = \begin{bmatrix} C \\ i \\ d \end{bmatrix}$$

Cipher ~~text~~ = Pohcid

Example 2

$$\text{Key} \rightarrow \begin{bmatrix} 3^+ & 10^+ & 26^+ \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix}$$

ciphertext $\rightarrow$ ^{24 0 9 12 6 22} ya'jmgw

① determine $\rightarrow 3(9 \times 17 - 4 \times 17) - 10(20 \times 17 - 9 \times 17) + 20(20 \times 4 - 9 \times 9) =$
 $(3 \times 85) - (10 \times 187) + 20 \times -1$
 $255 - 1870 - 20 = -1635 \% 26 = \boxed{3}$

② inverse of d $\rightarrow \boxed{9} \times 3 \% 26 = 1$

③ $\text{Key}^{-1} = \frac{1}{d} \text{adj}$

$$\text{adj} \rightarrow \begin{bmatrix} 3 & 10 & 20 \\ 20 & 9 & 17 \\ 9 & 4 & 17 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 10 & 20 & 3 & 10 \\ 20 & 9 & 17 & 20 & 9 \\ 9 & 4 & 17 & 9 & 4 \\ 3 & 10 & 20 & 3 & 10 \\ 20 & 9 & 17 & 20 & 9 \end{bmatrix} \rightarrow \begin{bmatrix} 9 & 17 & 20 & 9 \\ 4 & 17 & 9 & 4 \\ 10 & 20 & 3 & 10 \\ 9 & 17 & 20 & 9 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} 85 & -187 & -1 \\ -90 & -129 & 78 \\ -10 & 349 & -173 \end{bmatrix}^t \rightarrow \begin{bmatrix} 85 & -90 & -10 \\ -187 & -129 & 349 \\ -1 & 78 & -173 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 7 & 14 & 16 \\ 21 & 1 & 11 \\ 25 & 0 & 9 \end{bmatrix}$$

$$\text{Key}^{-1} = 9 \times \begin{bmatrix} 7 & 14 & 16 \\ 21 & 1 & 11 \\ 25 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 63 & 126 & 144 \\ 189 & 9 & 99 \\ 225 & 0 & 81 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 11 & 22 & 14 \\ 7 & 9 & 21 \\ 17 & 0 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 11 & 22 & 14 \\ 7 & 9 & 21 \\ 17 & 0 & 3 \end{bmatrix} \begin{bmatrix} 24 \\ 0 \\ 9 \end{bmatrix} = \begin{bmatrix} 390 \\ 357 \\ 435 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 8 \\ 19 \\ 19 \end{bmatrix} = \begin{bmatrix} a \\ t \\ t \end{bmatrix}$$

$$\begin{bmatrix} 11 & 22 & 14 \\ 7 & 9 & 21 \\ 17 & 0 & 3 \end{bmatrix} \begin{bmatrix} 12 \\ 6 \\ 22 \end{bmatrix} = \begin{bmatrix} 572 \\ 600 \\ 270 \end{bmatrix} \xrightarrow{\text{mod } 26} \begin{bmatrix} 0 \\ 2 \\ 10 \end{bmatrix} = \begin{bmatrix} a \\ c \\ k \end{bmatrix}$$

plaintext $\rightarrow$ Attack

Steps For adjoint:-

- 1) Repeat First two columns
- 2) Repeat First two rows
- 3) Remove First row and First column

All inverses:-

- 1) $1 \leftrightarrow 1$
- 2) $3 \leftrightarrow 9$
- 3) $5 \leftrightarrow 21$
- 4) $7 \leftrightarrow 15$
- 5) $11 \leftrightarrow 19$
- 6) $17 \leftrightarrow 23$
- 7) $25 \leftrightarrow 25$